

TetraTHON 2026

36-Hour Indo-French AI Hackathon

Navrachana University, Vadodara · July 31 – August 2, 2026

PRESCREENING ROUND

PROBLEM STATEMENT BOOKLET

Domains: AgriTech · EdTech · FinTech · HealthTech

Powered by India AI Mission · In collaboration with ISEN Méditerranée

AI-Powered Health Companion — Symptom Triage & Chronic Disease Self-Management

PROBLEM ABSTRACT

Access to timely primary healthcare and self-management support remains a critical challenge across semi-urban and rural India. Patients frequently present at emergency facilities with non-urgent conditions or delay care for genuinely serious ones, while those managing chronic diseases — Type 2 diabetes, hypertension, asthma — suffer from information overload, lack of personalisation, and poor adherence to self-management protocols. Participants must build an integrated AI-powered health companion that (a) accepts patient-entered symptoms and optional basic vitals to return a structured triage recommendation (home care / consult within 48 hours / proceed immediately to facility) with plain-language rationale and red-flag highlighting, and (b) tracks daily patient-reported metrics for one chronic condition, identifies adverse trends, delivers contextual nudges, and auto-generates a shareable weekly clinician summary — all operable in low-bandwidth conditions with a privacy-first data architecture.

EXPECTED OUTCOMES

- Symptom intake interface (web or chat) with optional vitals input and three-tier triage decision engine
- Plain-language rationale with red-flag highlighting and confidence/uncertainty indicator per decision
- Daily metric logging for one chronic condition with trend detection and contextual nudge/alert system
- Auto-generated weekly PDF/HTML clinician summary with privacy-first local or consented backend storage
- Mocked local resource/referral directory integration
- Optional: multilingual support (Hindi / regional language) and voice or WhatsApp-style chat interface

EVALUATION CRITERIA

Evaluation Criteria	Description
Understanding of the Problem Statement	Clear articulation of the problem, its context, and the pain points it addresses.
Proposed Approach	Methodology, intended tech stack, workflow, and reasoning behind design decisions.
Market Fit & Real-World Relevance	How the prototype solves a tangible real-world problem, the target audience it serves, and what makes the solution viable, scalable, and impact-driven.

Transparent Credit Scoring & AI-Driven Micro-Investment Advisor for Underserved Users

PROBLEM ABSTRACT

Over 190 million adults in India remain credit-invisible, lacking formal credit history, while millions more — especially first-time retail investors from Tier-2 and Tier-3 cities — face information asymmetry, jargon barriers, and absence of personalised guidance for small-ticket investments (Rs. 500–Rs. 5,000/month). Participants must build an integrated FinTech prototype that (a) designs a credit-likelihood scoring engine ingesting non-traditional digital signals (mobile recharge frequency, utility payment regularity, e-commerce transaction patterns) to produce an interpretable score with top-3 feature explanation and actionable improvement pathway, and (b) profiles a user's risk appetite through a short conversational assessment, maps their profile to suitable instrument categories, and generates a plain-language micro-investment allocation recommendation with a simulated 1–5 year projected growth chart — all built on synthetic or fully consented sample data with a clear disclaimer that outputs are for educational purposes only and do not constitute regulated financial advice.

EXPECTED OUTCOMES

- Feature engineering pipeline for non-traditional digital signals with scoring model and interpretability layer
- Risk bucket classification (Low / Medium / High) with top-3 feature explanations and score-improvement recommendations
- REST API endpoint (Flask/FastAPI) and dashboard UI showing at least 10 sample user profiles across all risk buckets
- 5–8 question conversational risk-profiling assessment with profile-to-instrument mapping engine
- Plain-language micro-investment recommendation with simulated growth projection chart (3 scenarios)
- Prominent disclaimer on all investment outputs: for educational purposes only, not regulated financial advice

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Adaptive Microlearning Engine & AI Doubt-Resolution Tutor for Mixed-Ability Classrooms

PROBLEM ABSTRACT

Indian classrooms routinely serve students spanning multiple learning levels within a single grade, yet most digital learning platforms deliver one-size-fits-all content and lack offline capability. Simultaneously, unresolved doubts in STEM subjects compound over time — evaluator-to-student ratios make real-time doubt resolution impractical, and generic chatbot tutors fail to identify the underlying conceptual gap driving a question. Participants must build an integrated EdTech prototype that (a) diagnoses a student's level via a 5-question probe, assigns one of three differentiated microlearning paths for a chosen grade-subject, delivers a 10-minute personalised micro-lesson with embedded practice problems, and provides an evaluator with a real-time cohort dashboard — deployable in low-bandwidth or offline environments — and (b) accepts a student's doubt (text or handwritten photo), identifies the root concept gap from a subject taxonomy, delivers a step-by-step Socratic or direct explanation, and maps the resolved doubt onto a concept graph to surface related weak areas.

EXPECTED OUTCOMES

- 5-question diagnostic probe with automatic level classification (Foundational / Grade-Level / Advanced)
- Three differentiated content paths with 10-minute micro-lessons, 3 embedded practice problems, and instant feedback per path
- Evaluator dashboard: per-student level, completion rate, and common error patterns; offline-capable or low-bandwidth-friendly architecture
- Question intake via text and optional image (handwritten problem photo) with root concept-gap identification
- Step-by-step explanation engine (Socratic or direct) with concept graph visualisation of resolved and related weak-area nodes
- Session history with mastery progression indicators per concept node; optional LLM-powered explanation with curriculum citation

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Precision Crop Advisory System & Post-Harvest Loss Reduction Planner for Smallholder Farmers

PROBLEM ABSTRACT

India's smallholder farmers — operating plots under 2 hectares and accounting for over 85% of the farming community — receive the least personalised advisory support, while India loses an estimated 15–20% of its agricultural produce to post-harvest deterioration driven by poor storage decisions, delayed market linkage, and lack of real-time price intelligence. Participants must build an integrated AgriTech prototype that (a) ingests simple farmer-provided inputs (field location, crop type, sowing date, recent weather observation, and an optional crop/leaf photo) and returns three ranked, actionable advisories for the next 7 days covering irrigation scheduling, fertiliser application, and pest or disease alert with confidence indicators and farmer-accessible plain-language rationale, and (b) accepts crop type, quantity, current storage condition, and farmer location to recommend an optimal sell/store/transport action using mocked or real mandi price data, estimated spoilage curves, and transport cost models — with an optional SMS/WhatsApp price-threshold alert simulation.

EXPECTED OUTCOMES

- Multi-input intake form: location (pin/GPS), crop type, sowing date, weather observation, optional leaf photo
- 3 ranked advisories (irrigation / fertiliser / pest alert) for next 7 days with confidence indicators (High / Medium / Low) and plain-language rationale
- Optional: integration with open weather API (OpenWeatherMap / IMD) and leaf disease classifier using lightweight CNN
- Crop + storage + location intake with spoilage-risk classifier and Sell / Store / Transport decision engine with expected return estimate
- Mocked or real mandi price feed (Agmarknet or synthetic) with spoilage curve visualisation and transport cost model
- Optional: SMS/WhatsApp alert simulation for price threshold triggers; prototype covers at least 4 crop types and 5 location combinations

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IMPORTANT NOTICE — PRE-SCREENING ROUND

As part of the pre-screening round, each team is required to submit a document covering the following:

1. **Understanding of the Problem Statement:** A clear articulation of the problem, its context, and the pain points it addresses.
2. **Proposed Approach:** Your methodology, intended tech stack, workflow, and the reasoning behind your design decisions.
3. **Market Fit & Real-World Relevance:** How your prototype solves a tangible real-world problem, the target audience it serves, and what makes your solution viable, scalable, and impact-driven.

Kindly ensure that your submission is a well-structured and original **PPT Document**, reflective of your team's collective thinking.

P.S. — GitHub Submission (Highly Recommended)

It would be highly beneficial if your team additionally submits an early prototype via a public GitHub repository, with regular and visible commits throughout the pre-screening period. Consistent progress on GitHub demonstrates your team's seriousness, accelerates your ideation-to-execution cycle, and gives the evaluation panel deeper insight into your problem-solving approach.

The date and time for the online evaluation will be shared with you separately via email. Our team will review your submission and notify the application status via email.

For any queries, feel free to reach out to the organizing team.

◆ Teams selected in the prescreening round will receive their full hackathon problem statement at the TetraTHON 2026 Inauguration Ceremony — July 31, 2026 at Navrachana University. ◆